MILITARY SPECIFICATION

BOMB, CHEMICAL, MARK 116 MOD 0

(WITHOUT FUZE AND BURSTER)

FILLING ASSEMBLY AND MARKING

1. SCOPE

1.1 This specification covers the filling of a bomb (weteye) with chemical agent GB.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-10758 - Chemical Agent, Isopropyl Methylphosphono-fluoridate (Fluoroisopropoxymethylphosphine Oxide).

STANDARDS

FEDERAL

FED. STD. 595 - Colors.

FSC 1325

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD270365 - Bomb, Chemical, Mark 116 Mod O - Filling Assembly.

LD270366 - Bomb, Chemical, Mark 116 Mod O - Metal Parts Assembly.

(All drawings referenced in this specification are listed on Dwg LD270366 or LD270365.)

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain that all materials and components listed on LD270366 and LD270365 have met their respective specifications prior to their use.

3.2 Filling and closing. The bomb shall be filled with GB (see 6.3) conforming to MIL-C-10758 and closed as shown on Drawing D1568458. From the time of filling until completion of test specified in 4.6.1, the bomb shall be kept in a nose down position (see 6.2).

3.3 Helium leakage. The bomb shall not leak helium through the closure at a rate greater than 1×10^{-6} atmospheric cubic centimeter per second when tested as specified in 4.6.1.

3.4 Agent contamination. The exposed surfaces of the bomb shall be free from agent contamination when tested as specified in 4.6.2 (see 6.5).

3.5 Filling level. The filling level shall be within the applicable limits shown in table I when tested as specified in 4.6.3.

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Table I - Filling level

Temperature range °F	Void height - *h (inches)	
	Minimum	Maximum
40 to 50	11.50	13.15
50 to 60	10.94	12.60
60 to 70	10.38	12.04
70 to 80	9.79	11.45
80 to 90	9.22	10.88
90 to 100	8.63	10.32
100 to 110	8.06	9.79

*h = Distance measured from the base of the bomb shell assembly, when the bomb is in a vertical position with the nose down, prior to reinstallation of the tail assembly on the shell.

3.6 Workmanship. The bomb shall be free from visible damage such as dents, cracks or deformation that could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will insure equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.1.1 Filling level. When an alternative procedure is approved for use in determining the filling level, the procedure shall be monitored in accordance with 4.6.3 to insure that the requirements of this specification are satisfied.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Substation inspection. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the bombs filled by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of agent shall be represented in any one lot of finished bombs.

4.4 Sampling for examination and tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.5 Inspection provisions.

4.5.1 Examination and tests. Sample bombs and the level A preparation of delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.5.2 Critical defects. Each bomb in the lot shall be inspected for critical defects listed in the classification of defects.

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4.5.3 Classification of defects.

4.5.3.1 Bomb, Chemical, Mark 116 Mod 0 (Dwg D1568458).

Categories	Defects	Acceptance standards
Critical:		
1	Helium leakage	4.6.1
2	Contamination	4.6.2
3	Filling level	4.6.3
Major:	AQL 1.0 percent defective	
101	Component incorrectly assembled	
102	Lifting plugs incorrect	
103	Marking incorrect	
104	Filler plug assembly incorrect	
105	Workmanship (3.6)	
Minor:	AQL 2.5 percent defective	
201	Marking illegible	
202	Filling insufficient	4.6.3

4.5.3.2 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Marking illegible
102	Marking incorrect
103	Identification plates missing
104	Swing bolt nuts torque incorrect
105	Detection plug missing

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4.6 Tests.

4.6.1 Helium leakage. Prior to closing, the void above the agent shall be evacuated to a pressure not greater than 5 pounds per square inch absolute (psia) and filled with helium to a pressure of 15 psia to insure a partial pressure due to helium of no less than 10 psia after closing. The closure area shall be sealed off by a test head manifolded to a vacuum system and a mass spectrometer type helium leak detector. The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during the test. The leak detector shall be calibrated to respond to a leak of 1×10^{-6} atmospheric cc per second of helium at the test head when a standard leak is introduced into the system at that point.

4.6.2 Bomb contamination. Contamination shall be determined by use of an approved Edgewood Arsenal GB Agent detector system.

4.6.3 Filling level. The height of the GB in the filled bomb shall be determined radiographically.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. The filled bomb shall be packed for shipment in a container as shown on Drawing D1516671.

5.2 Marking. In addition to any special marking required by the contract or order the container shall be marked in accordance with Drawing D1516672. Color code identification shall be as shown in figure 1 and Fed. Std. No. 595.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The Government will furnish leak tester, radiographic equipment and agent detector system when required.

6.2 Filling and closing. Care should be taken to prevent wetting the interior closure with GB and thereby making void the results of the helium test.

6.3 Cuation. Due to the hazards attending the filling of munitions with GB, suppliers should take precautions to protect personnel from injury during filling and handling.

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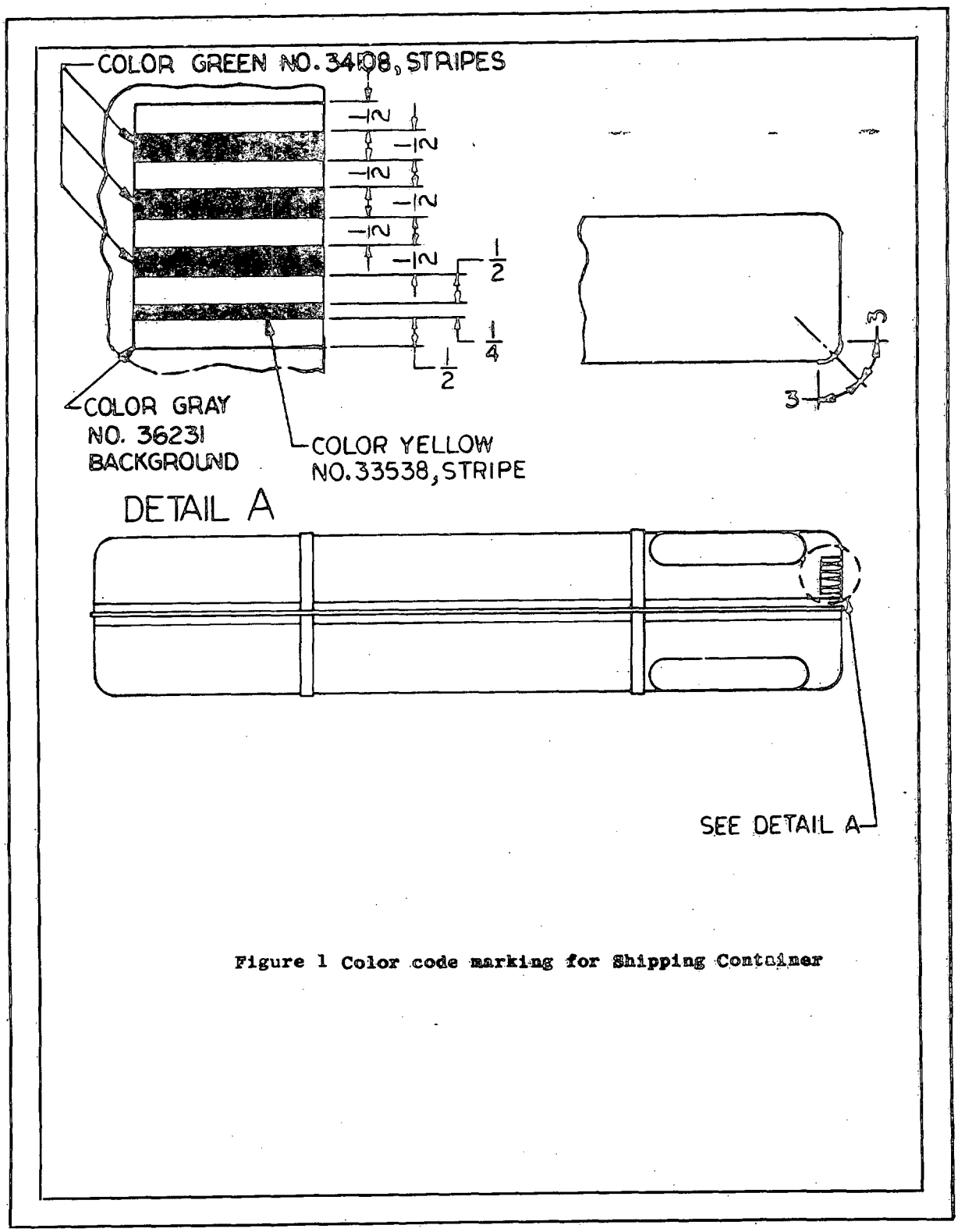


Figure 1 Color code marking for Shipping Container

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6.4 Filling aid. When filled as specified in 3.2 the bomb carries approximately the weight of GB specified on Drawing D1568458. This information is furnished as an aid to procurement and not as a criterion for filling.

6.5 Decontamination. Decontamination can be accomplished by subjecting the bomb to a spray of decontaminating solution (5 to 10 percent commercial grade soda ash Na_2CO_3) for a minimum of 1 minute and applied at a minimum pressure of 15 psig to completely wash all surfaces. The bomb should then be rinsed with water to remove all traces of the solution. The temperature of the spray and rinse water should be maintained at 140° to 160°F. Spray line should have a gage with an accuracy of 1 psig. An approved procedure of control of the solution strength should be maintained by periodic examination and the time the bomb is subjected to the spray should be automatically controlled and recorded.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1325-A140

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